ALCHEMY AOE ALLIANCE

TYPE: METHOD (MTHD)		**PROJECT: N/A**
MTHD--005		**REVISION: OI**
Selection, Round Robin Winners		**Page 1 of 8**

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1 - Definitions

The following mathematical definitions apply to player statistics within the same Round Robin tournament:

- ΔE – Elo Difference, Average Weighted. A value calculated by subtracting the tournament elo of opponents from tournament elo of a player, and then averaged to weigh number of games played against each. For example, if one game is played against an opponent 100 elo higher, and three games are played against an opponent 50 elo lower, then the average weighted elo difference would be: $\Delta E = \frac{(n_1 \Delta E_1 + n_2 \Delta E_2)}{(n_1 + n_2)} = \frac{(1)(-100)+(3)(+50)}{1+3} = \frac{-100+150}{4} = +12.5$.
- r – Win Rate, Actual. A value from 0 to 1, calculated as the quotient of games won and total games played.
- R – Win Rate, Expected. A value calculated from a player's tournament elo and the average tournament elo of their opponents. Follows the formula: $R = (1+10^{\frac{-\Delta E}{400}})^{-1}$. For example, a player with opponents on average 200 elo higher would have an expected win rate: $R = (1+10^{\frac{-(-200)}{400}})^{-1} = (1+10^{1/2})^{-1} = \frac{1}{(1+\sqrt{10})} = 0.24$ or 24%.
- p – Participation Rate. A value from 0 to 1, calculated by subtracting administrative penalties from the maximum allowable penalties, then dividing by that maximum. For example, a single administrative penalty where ten is the maximum results in a 90% participation rate, $p=0.9$. Unless otherwise specified, the maximum number of administrative penalties is equal to the maximum number of games planned for players.
- α – Participation Bias. A value optionally defined by tournament host in a handbook or equivalent, used to magnify or shrink the importance of Participation Rate p . If this factor is not defined, then it may be assumed to be 1.

2 - Overview

This document specifies a method for mathematical evaluation of player statistics in a Round Robin style tournament, to determine which are highest performing and eligible to receive monetary prize winnings.

3 - Background

The original document describing player compensation, MTHD-001, has been divided into two parts and is now deprecated. The first (this specification) defines how winning players are selected.

4 - Formulation

4.A - Tournament Grade

Tournament Grade G is an amalgamation of the contributing variables defined in §1, satisfying Equation 4.A below:

$$G = (\frac{r}{R})(p)^{\alpha}$$

The relationship of any one player's Tournament Grade to the Grades of all other players determines their eligibility to receive a fraction of the prize pool.

The factors on the right hand side of Equation 4.A are not exhaustive, but if a tournament would include a more complex score calculation, then its handbook will define additional factors to multiply in.

4.A.1 - Example

A Round Robin player has, a 48% win rate, and participation score of 92%. Calculate Tournament Grade if the theoretical win rate of that player was 37% given average skill of opponents.

Since bias factor was not mentioned, it is assumed as the default value defined in §1:

Participation Bias (α) = 1

From the problem statement, additional variables are identified:

Win Rate, Actual (r) = 0.48

Win Rate, Expected (R) = 0.37

Participation Rate (p) = 0.92

The remaining step is to calculate Tournament Grade:

$$G = \left(\frac{r}{R}\right)(p)^{\alpha} = \left(\frac{0.48}{0.37}\right)(0.92)^{1} = 1.1935$$

4.A.2 - Example

Player1 is in a close contest for achieving highest Tournament Grade against a friend participating in the same Round Robin (Player2), but incurred one administrative penalty in the first week. Assuming no additional penalties, determine the relative win rate Player1 would need in order to match Player2's Grade. Note that this tournament can have a maximum of 16 penalties, and Player2's relative win rate is 1.05.

Relative win rate is taken to be the actual win rate divided by the expected win rate, and is represented as $\bar{r} = \frac{r}{R}$, with $\bar{r}_1 = \frac{r_1}{R_1}$ representing Player1 and $\bar{r}_2 = \frac{r_2}{R_2}$ representing Player 2. Equation 4.A can be simplified thusly:

$$G = \left(\frac{r}{R}\right)(p)^{\alpha} = \bar{r}(p)^{\alpha}$$

Each player gets their own version:

$$G_1 = \bar{r}_1 (p_1)^{\alpha}$$

$$G_2 = \bar{r}_2 (p_2)^{\alpha}$$

To achieve the same Tournament Grade, $G_1 = G_2$:

$$\bar{r}_1 (p_1)^\alpha = \bar{r}_2 (p_2)^\alpha$$

Rewritten:

$$\bar{r}_1 = \bar{r}_2 \left(\frac{p_2}{p_1}\right)^\alpha$$

The participation bias is presumed 1, and with other variables inserted:

$$\bar{r}_1 \;=\; \bar{r}_2 \left(\frac{p_2}{p_1}\right)^\alpha \;=\; 1.05\left(\frac{1}{\left(\frac{15}{16}\right)}\right)^1 \;=\; 1.05\left(\frac{16}{15}\right)^1 \;=\; 1.12$$

Thus, in order to achieve the same Grade, Player1 would require a relative win rate of 1.12 to compensate for the administrative loss.

5 - Handicap

Expected Win Rate R must account for any handicap applied to players. For example, if a tournament "converted" a 1200 player into a 1400 player using handicap, then Expected Win Rate R would would consider the lower seeded player to be 1400, and not 1200 because that is the elo after modification.

5.A - Example

5.A.1 - Problem Statement

The performance of Player4 during a sample tournament is shown in the following table:

Higher Seed				***Lower Seed***				***Handicap***
Name	*Elo*	*Wins*	*Penalty*	*Name*	*Elo*	*Wins*	*Penalty*	
Player1	*1800*	*1*	*0*	*Player4*	*1300*	*2*	*0*	*10%*
Player2	*1700*	*2*	*0*	*Player4*	*1300*	*1*	*0*	*5%*
Player3	*1400*	*1*	*0*	*Player4*	*1300*	*1*	*1*	*0%*
Player4	*1300*	*2*	*0*	*Player5*	*1000*	*0*	*0*	*5%*

Table 5.A.1: Sample Tournament Results

Assume based on tournament methods that 10% handicap is worth 450 elo, while 5% is worth 200 elo across all skill ranges.

Perform the calculation of expected win rate for Player4 both with and without adjusting for handicap.

5.A.2 - Solution, Data Reformat

The first step in calculating ΔE is to simplify the relevant data provided in Table 5.A.1 as follows:

Player4 Elo		Opponent Elo		Elo Difference		Total Games
Tournament	Adjusted	Tournament	Adjusted	Tournament	Adjusted	
1300	1750	1800	1800	-500	-50	3
1300	1500	1700	1700	-400	-200	3
1300	1300	1400	1400	-100	-100	2
1300	1300	1000	1200	300	100	2

Table 5.A.2: Sample Tournament Reformatted Results

5.A.3 - Solution, Adjusted

Adjusted Elo Differences of Table 5.A.2 may be weighted by total games for each set using the formula found in §1, ΔE_{Adj} :

$$\Delta E_{Adj} = \frac{(n_1 \Delta E_1 + n_2 \Delta E_2 + n_3 \Delta E_3 + n_4 \Delta E_4)}{(n_1 + n_2 + n_3 + n_4)}$$

$$\Delta E_{Adj} = \frac{(3(-50)+3(-200)+2(-100)+2(100))}{(3+3+2+2)}$$

$$\Delta E_{Adj} = \frac{(-150-600-200+200)}{(10)} = -\frac{750}{10} = -75$$

Expected win rate calculated based on this:

$$R_{Adj} = \left(1+10^{\frac{-\Delta E_{Adj}}{400}}\right)^{-1} = \left(1+10^{\frac{-(-75)}{400}}\right)^{-1} = 0.3937$$

5.A.4 - Solution, Non-Adjusted

Tournament Elo Differences of Table 5.A.2 may then be weighted by total games for each set using the formula found in §1, ΔE_{NA} :

$$\Delta E_{NA} = \frac{(n_1 \Delta E_1 + n_2 \Delta E_2 + n_3 \Delta E_3 + n_4 \Delta E_4)}{(n_1 + n_2 + n_3 + n_4)}$$

$$\Delta E_{NA} = \frac{(3(-500) + 3(-400) + 2(-100) + 2(300))}{(3+3+2+2)}$$

$$\Delta E_{NA} = \frac{(-1500 - 1200 - 200 + 600)}{(10)} = -\frac{2300}{10} = -230$$

Expected win rate calculated based on this:

$$R_{NA} = \left(1 + 10^{\frac{-\Delta E_{NA}}{400}}\right)^{-1} = \left(1 + 10^{\frac{-(-230)}{400}}\right)^{-1} = 0.2102$$

Note that a “Non-Adjusted” elo difference is never used. Its calculation demonstrates its importance, since 0.21 is a little over half the correct value of 0.39.

6 - Cutoff

A Tournament Grade may be calculated for each tournament participant. Then the top “X” players may be selected for compensation, where “X” was defined beforehand by the handbook or equivalent. If this threshold is not specified, then it may be assumed to be the top 1/3 of tournament participants, rounded down.

7 - Conclusion

A model was presented for selecting prize winners of a Round Robin style tournament that can account for their performance (actual win rate), challenge level (expected win rate), and administrative burden (participation rate). An example was also provided showing the importance of elo adjustment for expected win rate when handicap is involved.

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